Fiscal Year:	FY 2008	**Task Last Updated:**	FY 10/08/2008
PI Name:	Dinges, David F. Ph.D.		
Project Title:	Countermeasures to Neurobehavioral Deficits from Cumulative Sleep Deprivation During Space Flight: Dose-Response Effects of Recovery Sleep Opportunities		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **Sleep**:Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	06/01/2004	**End Date:**	05/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	18
No. of Bachelor's Candidates:	41	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Banks, Siobhan (University of Pennsylvania Health System)		
Grant/Contract No.:	NCC 9-58-HPF00404		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	(1) The overarching goal of this project is to develop sleep schedule countermeasures to ensure optimal neurocognitive performance capability in astronauts during prolonged space flight. The primary aim is to determine the sleep dose-response effects of an acute change in sleep duration that occurs between two periods of chronic sleep restriction on neurocognitive performance functions, subjective states, and waking and sleep physiology. The optimal performance of astronauts during extended-duration space flight depends heavily on achieving recovery through adequate sleep. There is now extensive evidence that astronaut sleep in space averages 4 to 6.5 hours per day, and when critical operations (e.g., nighttime docking) are scheduled, very little sleep may be obtained during a day prior to the critical event. Ground-based experiments on healthy adults by our laboratory and others have demonstrated that limiting daily sleep duration to less than 7 hours leads to cumulative deficits in neurocognitive performance and alertness. Within 1-2 weeks of sleep restriction at levels experienced by astronauts, performance deficits were serious; impairments on tasks requiring sustained attention, working memory and cognitive throughput reached levels equivalent to those found after 1-2 nights of total sleep loss. The experiment will determine the countermeasure benefits for performance (during critical operations and subsequent days of sleep restriction) from an acute increase in sleep duration (i.e., single night of recovery sleep). In addition, generating sleep dose-response functions will provide critically needed information on the adverse performance consequences of an acute reduction in sleep duration below the chronic sleep-restriction level, which can occur in space flight prior to a day of critical operations. We will establish sleep dose-response curves for the immediate and delayed impact on neurobehavioral functions, of an acute (1 night) change in sleep duration midway in a period of chronic sleep restriction. We will determine if performance recovery is complete after 2 nights of extended sleep, following chronic sleep restriction. In addition to the impact of a single night intervention (specific aim 1), we seek to resolve whether complete neurobehavioral recovery from prolonged chronic sleep restriction is possible within 2 nights. We will investigate the relationship between sleep physiology and performance responses. We will investigate the effects of chronic sleep restriction, acute sleep intervention, and recovery sleep on cardiovascular indices. (2) We are currently in the process of performing preliminary analyses on the data collected. Specifically, we are analyzing the neurobehavioral performance changes across the experimental protocol, and the recovery phase. We are also beginning analysis for the construction of the dose response recovery curves from the chronic sleep restriction. We are in the process of manual scoring and analysis of the polysomnographic data. (3) At the current time we are still analyzing the data and we have not yet fully constructed the dose response recovery curves. Preliminary analysis however supports the hypothesis that as time in bed for sleep increases on the acute intervention night, following chronic sleep restriction, performance on the next day of simulated critical operations is improved in a sleep duration dose-response manner. (4) Data collection is complete and analysis of the data has commenced. We anticipate submitting several manuscripts to peer review journals this year, which will begin to reveal the relationship between the varying durations of time in bed and recovery of waking neurobehavioral and physiological outcomes, following chronic partial sleep deprivation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary aim is to determine the sleep dose-response effects of an acute change in sleep duration that occurs between two periods of chronic sleep restriction, on neurocognitive performance functions, subjective states, and waking and sleep physiology. The experiment will determine the countermeasure benefits for performance (during critical operations and subsequent days of sleep restriction) from an acute increase in sleep duration (i.e., single night of recovery sleep). The knowledge gained has the potential to change work scheduling and further understand the effect of sleep lose and recovery on neurobehavioral function in many Earth-based safety-sensitive occupations, such as transportation workers (e.g., truck drivers, train conductors, airline pilots); operators in safety-sensitive industries (e.g., power plant control rooms); and military personnel.
Task Progress:	Data collection is now complete. Seventy-two subjects completed the 16 day in-laboratory study protocol (for a total of 1152 laboratory days). We are currently in the process of performing analyses on the data collected. Specifically, we are analyzing the neurobehavioral performance changes across the experimental protocol, and the recovery phase. We are in the process of manual scoring and analysis of the polysomnographic and cardiovascular data. Preliminary data suggests that a single sleep period of 8h, 10h or 12h after 5 nights of 4h sleep, provides substantial acute recovery, but not protection against the immediate reappearance of performance deficits when it is followed by 5 additional nights of sleep restriction.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Abstracts for Journals and Proceedings	Avery N, Banks S, Dinges DF. "Change in psychomotor vigilance test lapses predicts change in digit-span memory performance during sleep restriction." Sleep 2007. 21st Annual Meeting of the Associated Professional Sleep Societies, Minneapolis, MN, June 9-14, 2007. Sleep. 2007;30 Suppl:A122. , Jun-2007
Abstracts for Journals and Proceedings	Avinash D, Banks S, Van Dongen H, Dinges DF. "Slow wave dynamics during consecutive weeks of sleep restriction to 4 hours per day." Sleep 2007. 21st Annual Meeting of the Associated Professional Sleep Societies, Minneapolis, MN, June 9-14, 2007. Sleep. 2007;30 Suppl:A119. , Jun-2007
Abstracts for Journals and Proceedings	Banks S, Bergamo C, Dinges DF. "Sleep restriction reduces heart rate variability." Sleep 2007. 21st Annual Meeting of the Associated Professional Sleep Societies, Minneapolis, MN, June 9-14, 2007. Sleep. 2007;30 Suppl:A31. , Jun-2007
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Abstracts for Journals and Proceedings	Caruso H, Banks S, Minkel J, Dinges DF. "Executive functioning following five nights of sleep restriction." Sleep 2007. 21st Annual Meeting of the Associated Professional Sleep Societies, Minneapolis, MN, June 9-14, 2007. Sleep. 2007;30 Suppl:A120. , Jun-2007
Abstracts for Journals and Proceedings	Erickson J, Banks S, Dinges DF. "Winscat test battery is partially sensitive to chronic sleep restriction." Sleep 2007. 21st Annual Meeting of the Associated Professional Sleep Societies, Minneapolis, MN, June 9-14, 2007. Sleep. 2007;30 Suppl:A124. , Jun-2007
Abstracts for Journals and Proceedings	Goel N, Lakhtman L, Basner M, Banks S, Dinges DF. "Neurobehavioral and cognitive differences during total versus partial sleep deprivation." Sleep 2007. 21st Annual Meeting of the Associated Professional Sleep Societies, Minneapolis, MN, June 9-14, 2007. Sleep. 2007;30 Suppl:A131. , Jun-2007
Abstracts for Journals and Proceedings	Goel N, Lakhtman L, Basner M, Banks S, Dinges DF. "Phenotyping neurobehavioral and cognitive responses to partial sleep deprivation." Sleep 2007. 21st Annual Meeting of the Associated Professional Sleep Societies, Minneapolis, MN, June 9-14, 2007. Sleep. 2007;30 Suppl:A130-1. , Jun-2007
Abstracts for Journals and Proceedings	Lakhtman L, Banks S, Goel N, Dinges DF. "Synthetic work performance following five nights of sleep restriction." Sleep 2007. 21st Annual Meeting of the Associated Professional Sleep Societies, Minneapolis, MN, June 9-14, 2007. Sleep. 2007;30 Suppl:A120. , Jun-2007
Articles in Peer-reviewed Journals	Banks S, Dinges DF. "Behavioral and physiological consequences of sleep restriction." J Clin Sleep Med. 2007 Aug 15;3(5):519-28. PMID: 17803017 , Aug-2007
Awards	Dinges DF. "1st Place Computational Modeling Prize for paper on 'Understanding Decrements in Knowledge Access Resulting from Increased Fatigue', The Cognitive Science Society, July 2007." Jul-2007
Awards	Dinges DF. "Laurence R. Young Space Biomedical Research Award for contributions to human performance in space, April 2008." Apr-2008
Awards	Dinges DF. "NASA Distinguished Public Service Medal, July 2007." Jul-2007
Books/Book Chapters	Mallis MM, Banks S, Dinges DF. "Sleep and circadian control of neurobehavioral functions." in "Neuroergonomics : the brain at work." Ed. R. Parasuraman, M. Rizzo. Oxford ; New York : Oxford University Press, 2007. p. 207-220., Jul-2007